

Strategically irrelevant framing governs cooperation in large language model agents

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Abstract

Large language models increasingly act as autonomous agents that negotiate, delegate and compete, yet we lack a principled account of what their behaviour in a social dilemma is. We report a fully crossed study of four frontier models playing 2,400 iterated prisoner’s dilemma dyads (48,000 binary decisions), crossing payoff magnitude, prompt language, assigned persona and replicate. Two manipulations that game theory guarantees are irrelevant nonetheless move behaviour: multiplying every payoff by a positive constant, which leaves the game strategically identical, shifts cooperation by up to 0.24, and changing the prompt language with objective framing held fixed shifts it by up to 0.15. A third, the assigned persona, moves cooperation by up to 0.35 while carrying no strategic information. Against this, the design explains only 18.7% of the variance in dyad-level cooperation: replicates of an identical prompt diverge more than the manipulations move the mean. The play is also not a strategy in the standard sense. Three of four models end games more cooperative than they start, contradicting backward induction, and no memory-one rule describes any model. Reporting a cooperation rate, or a canonical strategy label, without reporting framing sensitivity is not a measurement of the model.

Keywords: machine behaviour; large language models; iterated prisoner’s dilemma; cooperation; framing effects; evolutionary game theory

1 Introduction

The proposal that machines should be studied as behaving entities, with the observational tools of ethology and the experimental discipline of the behavioural sciences, rather than only as artefacts to be specified and verified, has become considerably more urgent since it was made [1]. Large language models now act as agents. They hold budgets, place orders, write and review code, negotiate on behalf of users, and increasingly encounter one another rather than a human counterpart. The population of interacting artificial agents is growing faster than our understanding of what governs their interaction, and the social dilemma is the canonical setting in which that question is posed.

The prisoner’s dilemma has served for four decades as the reference experiment for the emergence of cooperation among self-interested agents [2, 3]. Its appeal is that a very small amount of structure — a two-by-two payoff matrix with $T > R > P > S$ and $2R > T + S$ — generates a rich and well-mapped strategic landscape, in which reciprocity, forgiveness, retaliation

and exploitation can be defined precisely and their evolutionary fates computed [4–7]. That landscape underpins a large body of work on the statistical physics and evolutionary dynamics of cooperation [8–10], on the design of artificial agents that promote prosocial outcomes in mixed human–machine populations [11–13], and on mechanisms such as intention recognition and commitment that shift the cooperative equilibrium [14, 15].

It is therefore natural to ask where language models sit in that landscape, and a rapidly expanding literature does exactly this. Models have been probed with the instruments of cognitive psychology [16], compared against human participants across a battery of canonical games [17], used as simulated economic subjects [18–20], and placed directly into repeated dilemmas [21–23]. The resulting claims are typically of two kinds. The first is a scalar: model X cooperates at rate p . The second is a label: model X plays tit-for-tat, or is unforgiving, or is nicer than humans.

Both kinds of claim presuppose something that has not been established. A cooperation rate is a property of an agent only if it is stable under transformations of the problem that do not change the problem. A strategy label is informative only if the behaviour it names is actually generated by something in the family of strategies from which the label is drawn. Neither presupposition is safe for a system whose input is a natural-language string and whose behaviour is known to depend on the surface form of that string in ways that carry no task content [24–27]. Recent work has begun to make this concern concrete for strategic settings, showing that contextual framing can matter as much as game structure for model behaviour [28].

This paper takes the presupposition itself as the object of study. We adopt two invariance principles as the test. The first is the affine invariance of von Neumann–Morgenstern utility: if every payoff in a game is multiplied by the same positive constant, the resulting game is strategically identical, with the same best replies, the same equilibria and the same evolutionary dynamics [29, 30]. Any agent whose behaviour is described by a strategy over that game must be invariant to the transformation. The second is the invariance of a game to the language in which its rules are stated. A prisoner’s dilemma described in French is the same prisoner’s dilemma as one described in Chinese; a model that plays them differently is not responding to the game.

These two principles give us manipulations with an unusual property for behavioural work: the null hypothesis is not merely a convenient default but a theorem. If cooperation moves when payoffs are rescaled, the movement cannot be a strategic response, because there is no strategic difference to respond to. This converts a descriptive study into a test with a sharp prediction, and it is what allows the results below to be stated as failures rather than as observations.

We report a fully crossed factorial study in which four frontier language models play 2,400 iterated prisoner’s dilemma dyads, comprising 48,000 binary decisions. Payoff magnitude is crossed over three orders of magnitude, prompt language over five languages, and assigned persona over four pairings, with ten replicates in every cell. We ask three questions in sequence. Do the models satisfy the invariances? How much of their behaviour does the design explain at all? And is what remains describable as a strategy?

The answers are, in order, no, very little, and no. All four models violate at least one invariance, two of them substantially. The four design factors together account for 18.7% of the variance in dyad-level cooperation, so that repeated presentations of a byte-identical prompt disagree with one another more than the deliberate manipulations move the average. And the resulting play is not generated by a memory-one strategy: three of four models finish games

more cooperative than they begin, contradicting the backward-induction prediction for a finite horizon, and the nearest canonical rule is a poor description of the trajectory it names.

We do not read these results as showing that language models are bad at the prisoner’s dilemma. Optimal play is not the benchmark, and the models are not being asked to optimise. We read them as showing that the standard summary statistics of this literature — a cooperation rate, an archetype label — are not yet measurements of the model, because they are not stable under manipulations with respect to which any genuine measurement would have to be stable. The contribution is a diagnostic, and a demonstration that it changes what one is entitled to conclude.

2 Material and methods

2.1 Stage game and payoff geometry

Agents played a symmetric two-player prisoner’s dilemma with base payoffs $T = 10$, $R = 6$, $P = 2$, $S = 0$ (figure 1a). These satisfy the two defining constraints, $T > R > P > S$ and $2R > T + S$, so that mutual defection is the unique one-shot Nash equilibrium while mutual cooperation is Pareto superior to it. Two normalised descriptors of the dilemma’s severity locate this matrix within the family: greed, $(T - R)/(T - S) = 0.40$, is the incentive to defect against a cooperator, and fear, $(P - S)/(T - S) = 0.20$, is the incentive to defect against a defector.

Every cell of the matrix was multiplied by a payoff scale $\lambda \in \{0.1, 1, 10\}$, spanning three orders of magnitude. Because $\lambda > 0$, this is a positive affine rescaling: greed, fear, the ordering of outcomes, the best-reply correspondence and the equilibrium set are all unchanged, and the three games are the same game presented with different numbers.

The realised scale was recovered per game from the payoff values actually recorded rather than assumed from the directory in which the game was stored. This matters: 80 rounds (four dyads, 0.17% of the corpus) were found to carry $\times 1$ payoffs inside a folder nominally labelled $\times 10$, and are labelled by their realised scale in all sensitivity analyses.

2.2 Design

The design crossed four models — Claude 3.5 Haiku, Gemini 3.5 Flash-Lite, GPT-4o and Mistral Large — with three payoff scales, five prompt languages (English, French, Vietnamese, Chinese, Arabic), four persona pairings and ten replicates, giving 600 dyads per model and 2,400 in total (figure 1b). Each dyad ran for ten rounds and both agents are recorded, so the corpus contains 4,800 agent-games and 48,000 binary decisions. Every model $\times$ scale $\times$ language cell contains exactly 40 dyads; the design is balanced by construction, not by post-hoc weighting.

Each agent received a system-prompt persona, either *cooperative* or *selfish*, and the four pairings crossed the focal agent’s persona with its opponent’s. Critically, an agent was never told its opponent’s persona. The opponent’s persona is therefore a manipulation the focal agent can only learn about through play, which gives a clean separation between what the prompt does and what the interaction does.

2.3 Prompting protocol

Prompts were generated with the FAIRGAME framework [31], which supplies matched game templates in the five languages. On each round an agent received the full payoff table, its persona, the round index and the complete history of both players’ previous choices, and returned a single token naming one of two options. Communication between agents was disabled, so the only channel between them is the action history. Actions were labelled `OptionA` and `OptionB` in all languages, avoiding the connotations that words such as “cooperate” and “defect” would carry.

Decoding used a fixed low token budget with reasoning traces disabled. This detail is not cosmetic. Models that emit hidden reasoning before answering can exhaust a small output budget inside the trace, returning a truncated string that fails to parse; if the harness then falls back to a default action, the result is an apparent cooperation rate of 1.0 that is an artefact of the parser rather than a behaviour. We logged the parse-failure and fallback rates for every cell and required both to be below 0.02 before a cell was admitted.

2.4 Behavioural measures

The primary measure is the cooperation rate, the proportion of a game’s ten rounds on which an agent chose the cooperative action. Because both agents of every dyad appear as rows, the exploited (CD) and exploiting (DC) shares are equal by construction, and composition analyses merge them into a single anti-coordination band.

The memory-one fingerprint of an agent is the vector of four conditional probabilities $(p(C | R), p(C | S), p(C | T), p(C | P))$, the probability of cooperating after each of the four joint outcomes of the previous round. This is the complete description of a memory-one strategy and places every canonical rule at a known point: AllC at $(1, 1, 1, 1)$, AllD at $(0, 0, 0, 0)$, tit-for-tat at $(1, 0, 1, 0)$, win-stay lose-shift at $(1, 0, 0, 1)$ and grim trigger at $(1, 0, 0, 0)$. Two summaries are derived from it: niceness, the probability of cooperating on round one, and reciprocity, $p(C | C_{-1}) - p(C | D_{-1})$.

Payoff efficiency is the mean per-round payoff divided by R , the mutual-cooperation benchmark, which makes it comparable across λ .

2.5 Strategy read-out

Assigning a strategy label to observed play was done in two stages, which we keep distinct because they license very different claims.

The first stage is exact and deductive. For each agent-game we enumerated all 32 deterministic memory-one rules and retained those consistent with every transition actually observed. A game may be matched by exactly one rule, by several, or by none. Only the first case supports the statement that the agent played that rule.

The second stage is inductive and applies only where the first fails. A long short-term memory classifier [32] trained on synthetic play from known generating strategies returns a posterior over archetypes, and we take its argmax as the nearest neighbour. Its calibration was measured on held-out synthetic play against the Bayes ceiling implied by identical prefixes carrying different labels; that calibration describes the instrument, not the models, and we are explicit throughout about which of the two stages produced a given number.

We also recorded, for each agent-game, the minimum number of rounds on which play departs

from its own nearest rule. This is the quantity that tells one whether a label is a description or a gesture.

2.6 Statistical procedures

All intervals are 95% bootstrap confidence intervals over 2,000 resamples that resample whole dyads [33]. This is deliberate and consequential. Rounds within a game are strongly autocorrelated and the two agent-rows of a dyad share a history, so a round-level or row-level resample would understate the interval by roughly $\sqrt{2}$.

The payoff-scale slope is the change in cooperation rate per decade of λ , fitted by least squares on $\log_{10} \lambda$. Language disparity is the difference between a model’s highest- and lowest-cooperation language, tested against a permutation null that shuffles the language label across dyads within a model over 5,000 draws while holding the number of dyads per language fixed. Persona effects are differences of means with the same dyad-level bootstrap, reported alongside Cohen’s d .

The variance decomposition is sequential (Type I), entering factors in the order model, payoff scale, language, persona. Sequential decomposition credits shared variance to whichever factor is entered first, so entering model identity first makes its share an upper bound — the conservative direction for the argument we make from it.

Memory depth was compared by the Bayesian information criterion [34] across memoryless, memory-one, memory-two and memory-one-plus-phase specifications.

2.7 Reliability audit

Three checks were run whose expected answer is “nothing here”. Agents 1 and 2 of a matched pairing receive identical prompts, so their cooperation rates should agree; the observed correlation is $r = 0.67\text{--}0.95$ with a mean position bias of -0.017 , small but non-zero and carried into any claim that rests on a single role. The ten replicates of a cell share a prompt, and their spread is 1.3–1.6 times what independent rounds would produce, which is why every interval in this paper resamples dyads. Cell means reproduce across a random split of the dyads with Spearman–Brown reliability of 0.90–0.97, so the quantities we compare are stable even though individual games are not.

2.8 Two confounds declared in advance

Horizon. The Gemini arm was run with the round count disclosed to the agents; the Claude, GPT-4o and Mistral arms were run with the horizon hidden. A known finite horizon is precisely the condition under which backward induction predicts unravelling, so the Gemini column is “model $\times$ horizon” and not “model”, and it is marked as such in figure 1b. As shown below, this confound works against the conclusion we draw from it rather than for it.

Objective framing. The FAIRGAME templates do not state the agent’s objective identically in every language. The English, French and Vietnamese templates describe the outcomes as penalties and instruct the agent to minimise them; the Arabic and Chinese templates describe the outcomes as penalties but instruct the agent to maximise its rewards. Language is therefore confounded with objective framing in the five-language comparison, and we address this directly

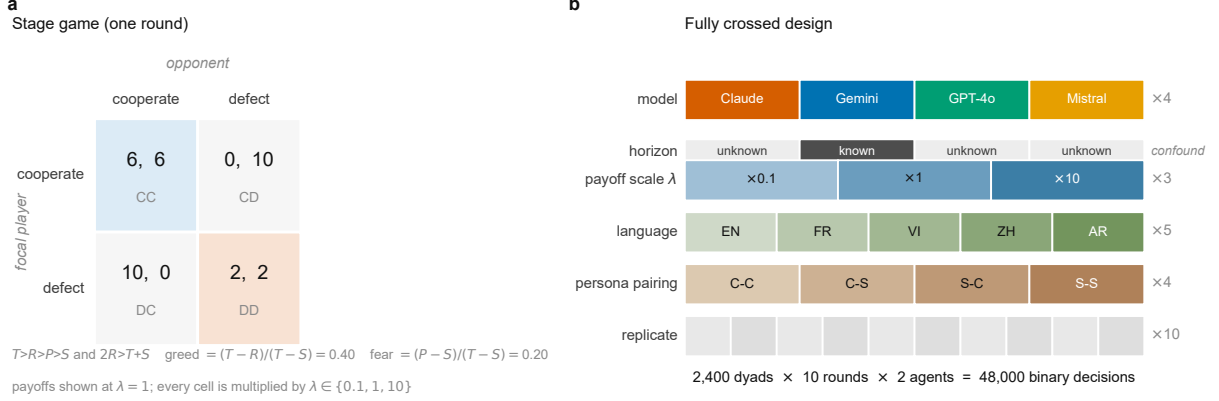


Figure 1: The game and the design. (a) The stage game at $\lambda = 1$. Payoffs satisfy $T > R > P > S$ and $2R > T + S$; greed = $(T - R)/(T - S) = 0.40$ and fear = $(P - S)/(T - S) = 0.20$ locate this matrix within the family of dilemmas. Every cell is multiplied by $\lambda \in \{0.1, 1, 10\}$, a positive affine rescaling that leaves the game strategically identical. (b) The fully crossed design. Four models × three payoff scales × five languages × four persona pairings × ten replicates = 2,400 dyads, 48,000 binary decisions, with 40 dyads in every model × scale × language cell. The horizon row is shaded as a confound rather than a factor: only the Gemini arm was told the round count.

in §3.3 by measuring the language effect within the three languages that share a single framing. We flag it here because it is a property of a widely used multilingual game-theoretic benchmark and not only of this study.

3 Results

3.1 Aggregate cooperation conceals more than it reports

Pooled across the entire design, the four models cooperate at very different rates: 0.55 for GPT-4o, 0.48 for Mistral Large, 0.38 for Claude 3.5 Haiku and 0.26 for Gemini 3.5 Flash-Lite. Read naively, this is the headline of a conventional benchmark report, and it is the form in which such results are usually communicated.

It is worth establishing immediately how little that number carries. The distribution of within-game cooperation is not concentrated near its mean for any model. For GPT-4o the mean of 0.55 falls in the emptiest region of its own distribution: the modal game is one in which the agent cooperates on every round, 14.9% of its games are entirely cooperative and 10.5% are entirely defecting. Mistral Large is more extreme still, with 24.3% of games entirely cooperative and 25.0% entirely defecting, so that the mean of 0.48 describes a bimodal mixture rather than a typical game. Gemini’s low mean of 0.26 is likewise produced substantially by a mass of 36.9% wholly defecting games rather than by uniformly guarded play.

The natural next question is how much of this spread the design accounts for. The answer is that it accounts for very little. In a sequential decomposition, model identity explains 9.5% of the variance in dyad-level cooperation, persona pairing 5.5%, payoff scale 3.6% and language 0.2%, for a total of 18.7%. The remaining 81.3% is within-cell: ten replicates of a byte-identical prompt disagree with one another more than all four deliberate manipulations together move the mean. Because model identity was entered first, its 9.5% is an upper bound.

This is the context in which the invariance tests should be read. We are not asking whether the models differ; we are asking whether the differences that manipulations produce are differences

Table 1: Behavioural profile of each model. Cooperation is pooled over the whole design. The payoff-scale slope is the change in cooperation per decade of λ , with 95% bootstrap CI; its null is exactly zero. The language gap is measured within the three languages sharing one objective framing. The persona effect is cooperative minus selfish. Distance is Euclidean, in the four-dimensional memory-one fingerprint space, to the nearest canonical rule; the canonical rules are at most 2.0 apart from one another. Exact rule is the share of agent-games in which exactly one of the 32 deterministic memory-one rules reproduces the whole trajectory.

Model	Coop. rate	Scale slope (per decade)	Lang. gap (fixed frame)	Persona effect	Nearest rule (dist.)	Exact rule
Claude 3.5 Haiku	0.38	-0.120 [-0.137, -0.102]	0.087**	+0.158	TFT (0.76)	12.5%
Gemini 3.5 Flash-Lite	0.26	-0.030 [-0.053, -0.007]	0.052	+0.045	GRIM (0.71)	40.8%
GPT-4o	0.55	-0.117 [-0.140, -0.094]	0.028	-0.018	GRIM (0.98)	17.3%
Mistral Large	0.48	-0.015 [-0.043, 0.012]	0.152***	+0.354	GRIM (0.94)	38.9%

** $P < 0.01$, *** $P < 0.001$ against a permutation null; unmarked gaps do not exceed their null.

a strategic agent could have. Table 1 summarises the per-model profile that the remainder of this section develops.

3.2 Payoff magnitude changes behaviour

Multiplying every payoff by a positive constant leaves best replies, equilibria and evolutionary dynamics unchanged. The null for a strategic agent is therefore not approximately zero but exactly zero.

Two of the four models violate it decisively (figure 2a). Claude 3.5 Haiku loses 0.120 cooperation per decade of λ (95% CI [-0.137, -0.102]) and GPT-4o loses 0.117 ([-0.140, -0.094]). Across the three scales this is a total swing of 0.24 and 0.23 respectively: GPT-4o cooperates at 0.71 when the same dilemma is presented with payoffs (1.0, 0.6, 0.2, 0) and at 0.47 when it is presented with (100, 60, 20, 0). Gemini shows a smaller but still non-zero slope of -0.030 ([-0.053, -0.007]), and only Mistral Large is consistent with invariance, at -0.015 ([-0.043, 0.012]).

The direction is consistent and interpretable: three of four models cooperate less as the stated stakes grow, even though the stakes are identical in every sense that a game-theoretic account can express. Note also that the effect is not monotone in the way a simple magnitude heuristic would predict. For Claude and GPT-4o almost the entire swing occurs between $\times 0.1$ and $\times 1$, with $\times 1$ and $\times 10$ nearly indistinguishable, which suggests sensitivity to the appearance of the numbers — fractional versus integral, single- versus multi-digit — rather than to magnitude as such.

Cooperation is not the only thing that moves. Payoff efficiency, measured relative to the mutual-cooperation benchmark, tracks the same pattern, and every one of the twelve model $\times$ scale cells sits above the efficiency a dyad would obtain if it reached the same cooperation rate without ever mistiming, by a premium of 0.033 to 0.088. The models earn more than coordinated play at their own cooperation rate would yield, and they do so through anti-coordinated rounds, which pay $(T + S)/2 = 5$ against $P = 2$ for mutual defection. Aggregate payoff efficiency is thus a poor proxy for cooperation here, and the two should not be used interchangeably.

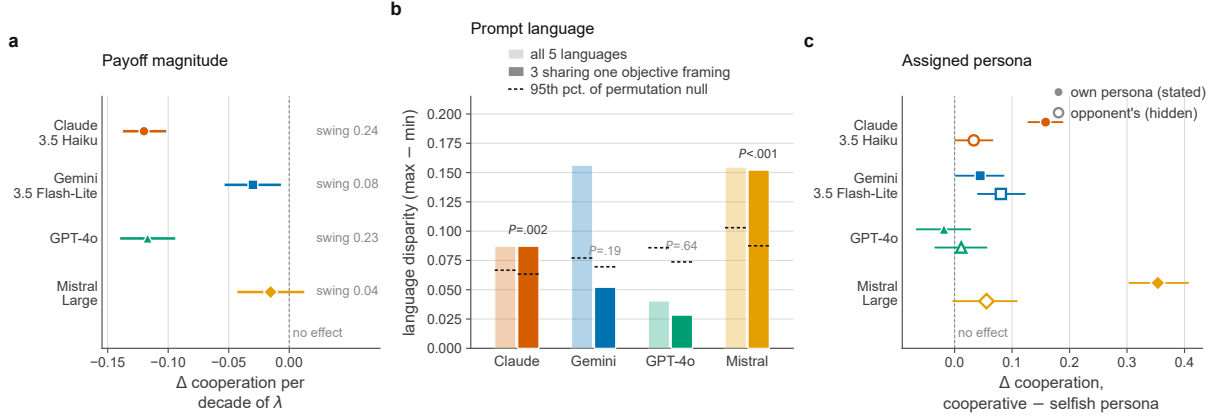


Figure 2: Three manipulations that cannot matter, and all three do. (a) Change in cooperation rate per decade of the payoff scale λ , with 95% bootstrap confidence intervals resampling whole dyads. A positive affine rescaling leaves the game strategically identical, so the null is exactly zero. Claude and GPT-4o depart from it by more than 0.11 per decade; the annotation gives the total swing across the three scales. (b) Language disparity, the gap between a model’s highest- and lowest-cooperation language. Pale bars use all five languages, in which objective framing varies with language; solid bars use only English, French and Vietnamese, which share a single framing. Dashes mark the 95th percentile of a permutation null shuffling the language label within a model over 5,000 draws; P values are for the framing-controlled comparison. (c) Effect of the assigned persona. Filled markers give the difference between an agent told it is cooperative and one told it is selfish; open markers give the same difference for the *opponent’s* persona, which is never disclosed and can only be inferred from play.

3.3 Prompt language changes behaviour, with framing held fixed

Measured across all five languages, three of four models show a language disparity beyond its permutation null: 0.156 for Gemini ($P < 0.001$), 0.155 for Mistral ($P < 0.001$) and 0.087 for Claude ($P = 0.001$), against 0.040 for GPT-4o ($P = 0.70$).

That comparison is confounded, and we treat the confound as a result rather than as a footnote. As set out in the methods, the English, French and Vietnamese templates instruct the agent to minimise a penalty, whereas the Arabic and Chinese templates instruct it to maximise a reward. Under the literal penalty-minimising reading, the dominance relation between the two options reverses relative to the reward-maximising reading, so a raw comparison across all five languages mixes a language manipulation with an objective-framing manipulation.

Two things follow. First, the models do not appear to be performing the dominance calculation under either reading. In every one of the five languages, agents told they are cooperative open with the same option far more often than agents told they are selfish, with a gap of 0.42 to 0.73; the action that functions as cooperative is the same one throughout, regardless of whether the stated objective is to minimise or to maximise. Models are tracking the semantic frame of the prompt rather than the arithmetic of the objective they were given.

Second, the language effect must be re-tested with framing held fixed. Doing so within English, French and Vietnamese alone (figure 2b, solid bars) leaves the effect intact for two of four models: Mistral Large retains a disparity of 0.152 ($P < 0.001$) and Claude 0.087 ($P = 0.002$), both above their permutation nulls. Gemini’s apparent disparity, by contrast, collapses from 0.156 to 0.052 and ceases to be significant ($P = 0.19$), identifying most of its five-language gap as a framing effect rather than a language effect. GPT-4o remains invariant on both readings ($P = 0.64$). The framing contrast itself is small and inconsistent in sign across models (-0.064 to $+0.093$), so it does not act as a simple main effect either.

The surviving effects are substantial. Mistral Large cooperates at 0.39 in English and 0.54 in Vietnamese — three languages, one objective framing, one payoff matrix, a gap of 0.15 in cooperation. We emphasise that this is now a claim about two of four models rather than three, and that the more cautious claim is the one the data support.

3.4 Assigned persona changes behaviour, and leaks

The persona manipulation differs from the previous two in that it is not strategically irrelevant by theorem; a persona could in principle be read as information about the kind of player one is facing. But an agent’s own persona carries no information about the game, and the four persona pairings present identical payoff matrices.

The effect is nonetheless the largest of the three (figure 2c). Mistral Large cooperates 0.354 more when told it is cooperative than when told it is selfish ($[0.303, 0.408]$, $d = 0.94$), and Claude 0.158 more ($[0.127, 0.189]$, $d = 0.61$). Gemini shows a small effect of 0.045 ($[0.001, 0.086]$, $d = 0.14$) and GPT-4o none at all (-0.018 , $[-0.067, 0.028]$). The effect is concentrated in the opening move: told it is cooperative, Mistral opens cooperatively in 81% of games; told it is selfish, in 0% of games. Claude moves from 86% to 19%, Gemini from 46% to 2% and GPT-4o from 62% to 16%.

Two observations complicate the reading that persona simply installs a disposition. The first is that the opening it installs is not profitable. For Mistral, opening cooperatively costs 1.82 payoff units per round ($[-2.15, -1.47]$) relative to opening with defection; for the other three models the difference is indistinguishable from zero. A persona that pushes an agent towards a costly opening and no compensating downstream advantage is behaving as a stylistic instruction rather than as a strategic commitment.

The second is that the opponent’s persona — never disclosed, inferable only from play — has a detectable effect for Gemini, which cooperates 0.080 more against a cooperative opponent ($[0.039, 0.123]$), and marginally for Claude (0.033 , $[0.000, 0.067]$). For Gemini this opponent effect is larger than its own-persona effect. That is the signature of an agent responding to the interaction rather than to the instruction, and it is the one place in this study where a model behaves more like a strategic player than like a prompt-follower.

3.5 The play is not a strategy

No unravelling. With a finite and commonly known horizon, backward induction predicts defection in every round; with an uncertain horizon the prediction weakens but the endgame effect remains the canonical observation in human play. Only Claude 3.5 Haiku shows it, declining from 0.53 in round one to 0.26 in round ten, a change of -0.27 (figure 3a). The other three models end *more* cooperative than they start: GPT-4o by $+0.20$, Gemini by $+0.08$ and Mistral by $+0.08$.

The horizon confound turns out to strengthen rather than weaken this observation. If unravelling were being suppressed simply because agents did not know when the game ended, the arm that did know should unravel. Gemini is that arm, and it warms up. Whatever prevents backward induction here is not ignorance of the horizon.

No memory-one rule. The memory-one fingerprint places every model far from every canonical strategy (figure 3b). The nearest archetype is tit-for-tat for Claude at Euclidean

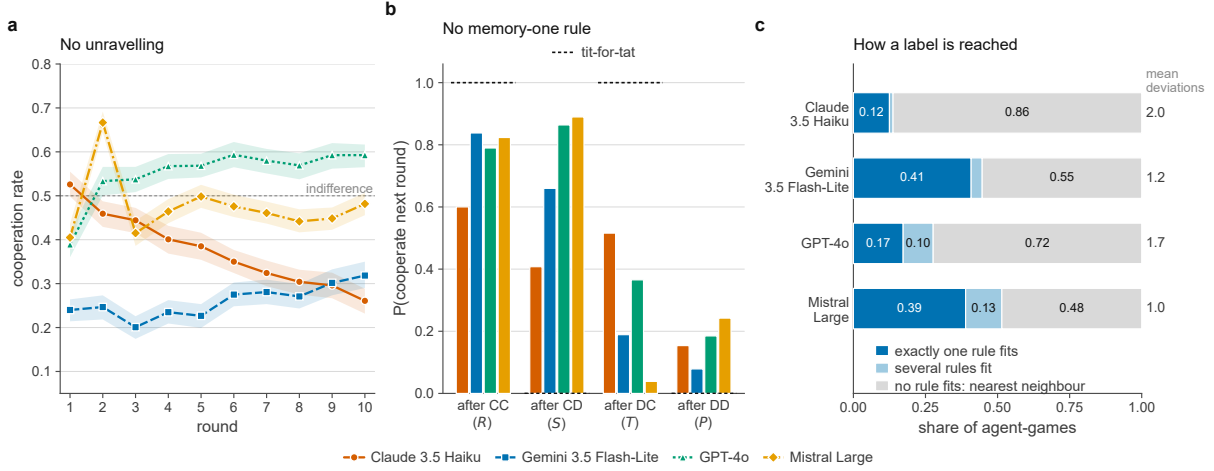


Figure 3: What the play is not. (a) Cooperation by round with 95% bootstrap bands. Backward induction from a finite horizon predicts decay towards mutual defection; only Claude decays. Gemini is the single arm that was told the round count, and it warms up rather than unravelling. (b) The memory-one fingerprint: probability of cooperating after each of the four joint outcomes of the previous round. These four numbers are the complete description of a memory-one strategy. Dashes mark tit-for-tat, which is 1 after CC and DC and 0 after CD and DD. No model approaches it: all four cooperate readily after being exploited (CD), where every canonical rule except AllC prescribes 0, and three of the four cooperate less after exploiting (DC) than after being exploited, which inverts reciprocity. (c) How an archetype label is actually reached. Dark: exactly one of the 32 deterministic memory-one rules reproduces the whole trajectory. Mid: several do, and the honest answer is the set. Grey: none does, and the label is a nearest-neighbour statement. The right-hand column gives the mean number of rounds out of ten on which play departs from its own nearest rule.

distance 0.76 in the four-dimensional fingerprint space, and grim trigger for Gemini (0.71), Mistral (0.94) and GPT-4o (0.98). For scale, the canonical rules are at most 2.0 apart from one another in this space, so a distance of 0.7–1.0 is not a near miss.

The shape of the departure is more informative than its size. All four models cooperate readily after having been exploited — $p(C | S)$ is 0.41 for Claude, 0.66 for Gemini, 0.86 for GPT-4o and 0.89 for Mistral — where tit-for-tat, grim and win-stay lose-shift all prescribe 0. And three of four cooperate *less* after having successfully exploited the opponent than after being exploited by it: $p(C | T)$ is 0.51, 0.19, 0.36 and 0.04 respectively, where tit-for-tat prescribes 1. Mistral Large’s reciprocity index is actually negative, -0.149 , meaning it is more likely to cooperate after the opponent defected than after the opponent cooperated. This is not a degraded reciprocator; it is a different object.

Bayesian information criteria confirm that memory-one is the wrong family rather than merely a poor fit. For all four models a memory-two specification improves on memory-one by a substantial margin, indicating dependence on history deeper than the previous round.

A label is mostly not a description. Combining these, we ask how often an archetype label is earned deductively rather than assigned by nearest neighbour (figure 3c). Exactly one of the 32 deterministic memory-one rules reproduces the entire trajectory in 12.5% of Claude’s agent-games, 17.3% of GPT-4o’s, 38.9% of Mistral’s and 40.8% of Gemini’s. In the remaining 86%, 72%, 48% and 55% respectively, no rule in the class fits and the label is a nearest-neighbour statement produced by the classifier. Where a label is assigned this way, play departs from the rule it names on 1.0 to 2.0 rounds out of ten.

A further caveat limits even the deductive fraction. Ten rounds rarely visit all four conditioning states, so rules that differ only in an unvisited state cannot be distinguished. Of the 4,800 agent-games, 2,192 match at least one of the 32 rules but only 89 match exactly one. Coverage figures for the wider class are therefore statements about a *set* of rules, not about a rule.

Widening the vocabulary helps, but not enough and not for free. Moving from the four canonical archetypes to all 32 memory-one rules and then to a one-switch two-regime model raises explained coverage to 0.80, 0.85, 0.73 and 0.58 for Gemini, Mistral, GPT-4o and Claude. But a shuffled null over the same classes already reaches much of that, and the excess over chance is 0.02 for the canonical four, 0.07–0.13 for the 32 memory-one rules and 0.20–0.24 for the two-regime model. Between 15% and 42% of agent-games remain unexplained by any of them.

3.6 What is stable

It would be a misreading to conclude that nothing here is measurable. The quantities we compare are reliable even though individual games are not. Split-half reliability of the cell means is 0.90 for Claude, 0.91 for GPT-4o, 0.95 for Gemini and 0.97 for Mistral. The models are also clearly distinct from one another in fingerprint space and in their response to every manipulation: the differences reported above are not noise.

What is unstable is the individual game. Ten replicates of an identical prompt have a standard deviation 1.3 to 1.6 times what independent rounds would produce, and within a dyad the two agents’ realised payoffs differ by a median of 5.0 units for Mistral and 3.0 for GPT-4o. The correct summary is that these systems have reproducible *propensities* and irreproducible *episodes*, and that a benchmark reporting the former as though it characterised the latter is over-claiming.

4 Discussion

4.1 What an invariance failure means

The central results are failures of two invariances that are not empirical regularities but consequences of how games are defined. Multiplying every payoff by a positive constant produces the same game; stating a game in a different language produces the same game. An agent whose behaviour is described by a strategy over a game must therefore be invariant to both. Two of four models move by more than 0.11 in cooperation per decade of payoff scale, and two of four move by 0.09–0.15 across languages that share an objective framing.

The consequence is not that the models are irrational in the ordinary sense of choosing dominated actions. It is stronger and more specific: the object that a cooperation rate is supposed to measure is not well defined for these systems, because the measurement changes when the thing measured does not. Reporting that a model cooperates at 0.55 without reporting that the same model cooperates at 0.71 or 0.47 depending on whether the payoffs are written as decimals or as hundreds is not a partial description; it is a description of an artefact of presentation.

This connects the machine-behaviour programme [1] directly to the framing literature in human decision making [35, 36]. Humans also violate invariance, and the discovery that they do reshaped economics. The disanalogy is instructive, though. Human framing effects are systematic, have been mapped to a compact set of psychological mechanisms, and are reproducible across

populations. What we observe here is heterogeneous across models — Mistral is scale-invariant but the most persona-steerable; GPT-4o is language-invariant and persona-inert but strongly scale-sensitive — and sits on top of a within-cell variance that dwarfs it. There is no single “LLM framing effect” to be characterised, only a per-model, per-manipulation profile that must be measured.

4.2 Consequences for strategy attribution

A large part of the appeal of the prisoner’s dilemma is that its strategy space is well mapped, and it is tempting to import that map wholesale when describing language models. Our results suggest the import is not licensed.

The memory-one fingerprints do not merely fail to coincide with canonical rules; they have a shape no canonical rule has. Cooperating readily after being exploited and reluctantly after exploiting is the reverse of the reciprocity that makes tit-for-tat evolutionarily interesting [2, 4]. Mistral Large’s negative reciprocity index is not a noisy zero but a systematic inversion. Whatever generates this behaviour, it is not a rule that conditions on the previous round in the way the memory-one family does, and the BIC comparison confirms that history dependence runs deeper than one round.

The practical recommendation is a reporting standard. A statement of the form “model X plays tit-for-tat” should be accompanied by the fraction of agent-games in which a rule was deductively established rather than assigned by nearest neighbour, and by the number of rounds on which play departs from its assigned label. On our corpus those numbers are 12–41% and 1.0–2.0 rounds out of ten. Without them, an archetype label reports the resolution of the classifier rather than the behaviour of the model.

4.3 Implications for evolutionary and hybrid populations

The results bear directly on the growing effort to model populations in which artificial and human agents interact [11–13]. Such models are typically built by assigning agents strategies from a known family and computing the resulting dynamics. Two of our findings complicate that programme.

First, if an artificial agent’s cooperation depends on the units in which payoffs are written, then its behaviour is not a function of the game and cannot be carried across a replicator equation that only knows the game. Two population models identical up to a rescaling of the payoff matrix would make different predictions about the same artificial agent, which is not a property any of these frameworks are built to accommodate.

Second, if 81% of the variance in cooperation is within-cell, then the appropriate representation of such an agent in a population model is not a strategy but a distribution over strategies, and possibly one whose support changes with framing. The absorbing behaviour we observe — 25% of Mistral’s games entirely defecting and 24% entirely cooperative — suggests that the relevant dynamics may be better captured by mixtures with absorbing states than by a single reactive rule. Whether the well-understood results on the evolution of reciprocity [5–7] survive that substitution is, we think, an open and interesting question, and one this theme issue is well placed to pose.

4.4 Limitations

Four limitations bound these conclusions, and we state them plainly.

The model set is the most consequential. Four models is a small sample of a fast-moving population, and three of the four belong to an earlier generation than the fourth. Nothing here establishes that the pattern generalises to models with explicit reasoning procedures, which may well perform the dominance calculation the present models appear to skip. Testing scale invariance on reasoning models is the single most valuable extension of this work, and we regard the present results as establishing the diagnostic rather than settling the question.

The horizon confound remains. The Gemini arm knew the round count and the others did not, so the Gemini column mixes model identity with horizon knowledge. We have argued that this works against our conclusion in the one place it matters — the arm that knew the horizon warms up rather than unravelling — but it would be better to cross horizon explicitly.

The objective-framing inconsistency in the multilingual templates is inherited from a widely used benchmark and, as reported above, absorbs most of one model’s apparent language effect. Our framing-controlled analysis rests on three languages rather than five, which reduces power; the two null results (Gemini, GPT-4o) should be read with that in mind. We would encourage users of multilingual game-theoretic benchmarks to verify objective statements across translations before treating language as a clean factor.

Finally, ten rounds is short. It under-identifies the memory-one vocabulary, as reported, and it compresses any endgame effect into a small number of rounds. A longer horizon would sharpen both the strategy read-out and the unravelling test.

4.5 Conclusion

Four frontier language models playing 2,400 iterated prisoner’s dilemmas fail invariances that any game-theoretic description of them would require. Rescaling payoffs, which cannot change the game, moves cooperation by up to 0.24. Changing the prompt language while holding the objective framing fixed moves it by up to 0.15. Assigning a persona, which carries no strategic information, moves it by up to 0.35. Against these, the entire crossed design explains 18.7% of the variance in dyad-level cooperation, and the play that results is not generated by any memory-one rule: three of four models end more cooperative than they start, three of four cooperate more after being exploited than after exploiting, and in half to seven-eighths of agent-games no rule in the class fits at all.

We therefore propose a minimal standard for this literature. A cooperation rate reported for a language model should be accompanied by its sensitivity to payoff scale and to prompt language, and a strategy label should be accompanied by the fraction of games in which that label was deductively established. These are cheap to compute and they change what the numbers mean. Until they are routine, the field is at risk of building an evolutionary game theory of artificial agents on quantities that are properties of prompts rather than of agents.

Data accessibility

The full corpus of 2,400 dyads, the analysis pipeline that produces every figure and table, and the 26-figure supplementary suite are available at [REPOSITORY URL]. Every number appearing in a figure is also written to a machine-readable CSV file in the same repository. A permanent

archive with a DOI will be deposited on acceptance.

Competing interests

The authors declare no competing interests.

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